



Capstone Project Report on

Bluestock Data Analytics Internship Program

MUTUAL FUND ANALYTICS & PERFORMANCE EVALUATION SYSTEM

An End-to-End Data Analytics Solution Using Python, SQLite & Power BI

Technologies Used

Python • Pandas • NumPy • SQLite • SQL • Jupyter Notebook • Power
BI • Git • GitHub

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Introduction

The Indian mutual fund industry has experienced unprecedented growth over the past decade, driven by increasing financial awareness, digital investment platforms, and the rapid adoption of Systematic Investment Plans (SIPs). Millions of retail investors now participate in financial markets through mutual funds, making data-driven investment analysis more important than ever. As the number of schemes, fund houses, and investment products continues to grow, investors require analytical tools capable of evaluating performance, understanding risk, and identifying suitable investment opportunities. Traditional methods of analysing mutual funds often rely on isolated performance indicators or static reports that fail to provide a comprehensive understanding of fund behaviour. Financial analysts increasingly rely on modern data analytics techniques to combine historical Net Asset Value (NAV), Assets Under Management (AUM), investor behaviour, benchmark performance, and portfolio composition into a unified analytical framework. This project addresses that need by developing a complete Mutual Fund Analytics and Performance Evaluation System. The solution integrates data engineering, statistical analysis, financial performance metrics, and business intelligence to transform raw mutual fund data into actionable insights. Beginning with automated data ingestion and cleaning, the project builds a structured SQLite database that serves as the foundation for advanced financial analytics. The analytical component evaluates mutual fund performance using widely accepted financial metrics such as Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, Tracking Error, Maximum Drawdown, Historical Value at Risk (VaR), and Conditional Value at Risk (CVaR). These measures enable comprehensive assessment of both returns and associated risks. Beyond traditional performance analysis, the project incorporates advanced analytical techniques including Rolling Sharpe Ratio analysis, Investor Cohort Analysis, SIP Continuity Analysis, portfolio concentration measurement using the Herfindahl–Hirschman Index (HHI), and a rule-based mutual fund recommendation system that suggests suitable funds based on investor risk appetite.

To make these analytical insights accessible to decision-makers, an interactive four-page Power BI dashboard was developed. The dashboard combines Key Performance Indicators (KPIs), trend analysis, comparative visualisations, heatmaps, and interactive slicers to enable dynamic exploration of the mutual fund ecosystem.

Overall, the project demonstrates how modern data analytics, financial modelling, and business intelligence tools can be integrated into a single decision-support platform capable of assisting investors, analysts, and financial institutions in evaluating mutual fund performance effectively.

Dataset Description and Technology Stack

1) Dataset Description

The success of any data analytics project depends on the quality, diversity, and completeness of the underlying datasets. This project utilizes ten carefully selected datasets that collectively provide comprehensive information about the Indian mutual fund industry. These datasets cover multiple aspects of mutual fund operations, including scheme information, historical Net Asset Value (NAV), Assets Under Management (AUM), investor transactions, portfolio holdings, benchmark indices, and industry-level statistics. Together, they provide a strong foundation for performing financial analysis, risk assessment, investor behaviour analysis, and dashboard visualisation. Prior to analysis, each dataset underwent an extensive preprocessing stage to ensure consistency and accuracy. Missing values were identified and handled appropriately, duplicate records were removed, inconsistent column names were standardised, and date fields were converted into a uniform format. This preprocessing step ensured that the datasets could be integrated seamlessly for further analysis and eliminated issues that could affect the reliability of the analytical results.

Dataset	Purpose
Fund Master	Stores information about mutual fund schemes including fund house, category, benchmark, fund manager, and investment requirements.
NAV History	Contains historical daily Net Asset Values used for return calculations and financial performance analysis.
AUM by Fund House	Provides Assets Under Management data for different Asset Management Companies to analyse market share and industry growth.
Monthly Inflows	SIP Records monthly Systematic Investment Plan (SIP) investments across the industry to study investment trends.

Category Inflows	Captures category-wise inflows and outflows to identify investor preferences among different mutual fund categories.
Industry Folio Count	Represents the number of investor accounts over time, indicating overall industry participation.
Scheme Performance	Includes historical returns over different investment horizons for comparing mutual fund performance.
Investor Transactions	Contains purchase, redemption, and SIP transaction details used for behavioural analysis and cohort identification.
Portfolio Holdings	Provides sector-wise allocation and stock holdings of mutual fund portfolios for concentration analysis.
Benchmark Indices	Contains historical benchmark index values such as NIFTY indices for evaluating fund performance against the market.

The project makes use of the following datasets: These datasets collectively enable a holistic analysis of the mutual fund ecosystem by combining operational, financial, and investor-related information. Their integration allows the project to evaluate fund performance from multiple perspectives rather than relying solely on historical returns.

2) Technology Stack

The implementation of this project required a combination of programming languages, analytical libraries, database technologies, and business intelligence tools. Python served as the primary programming language due to its extensive ecosystem for data analysis and financial computing. Pandas and NumPy were used for data manipulation and numerical operations, while SQLite was selected as the relational database because of its lightweight architecture and ease of integration with Python. Jupyter Notebook was used to perform exploratory data analysis, calculate financial metrics, and document the analytical workflow. Finally, Power BI was employed to build an interactive dashboard that enables users to explore fund performance, investor behaviour, and market trends through dynamic visualisations and slicers.

Table 2.1: Technology Stack Used

Technology	Role in the Project
Python	Data processing and analytics
Pandas & NumPy	Data cleaning and numerical analysis
SQLite	Structured data storage
Jupyter Notebook	Analysis and experimentation
Power BI	Interactive dashboard development
Git & GitHub	Version control and project management

ETL Pipeline and Database Design

The Mutual Fund Analytics & Performance Evaluation System follows a structured Extract, Transform, and Load (ETL) process to convert raw financial datasets into a clean, reliable, and analysis-ready database. Since the project utilizes multiple datasets collected from different sources, establishing an efficient ETL pipeline was essential to ensure consistency, improve data quality, and simplify downstream analytical tasks. The pipeline was developed entirely using Python and automated through custom scripts, reducing manual effort while improving reproducibility.

The ETL process begins with the extraction of ten raw CSV datasets containing information related to mutual fund schemes, historical NAV values, Assets Under Management (AUM), SIP inflows, category-wise investments, investor transactions, portfolio holdings, benchmark indices, and industry statistics. Each dataset is imported using the Pandas library, where an initial validation is performed to verify file availability, column names, data types, and overall dataset integrity before further processing.

Once the data has been extracted, the transformation stage focuses on improving data quality. Missing values are identified and handled using suitable techniques depending on the nature of the data. Duplicate records are removed to prevent incorrect calculations during analysis, while inconsistent column names are standardized to maintain a common schema across all datasets. Date columns are converted into a uniform datetime format, numerical values are validated to eliminate invalid entries, and categorical fields are cleaned to ensure consistency in naming conventions. These preprocessing operations significantly improve the accuracy of financial calculations and make it easier to combine information from multiple datasets.

After transformation, each cleaned dataset is stored separately inside the processed directory while preserving the original files in the raw directory. Maintaining separate raw and processed datasets ensures that the original source data remains unchanged, improves project reproducibility, and allows the cleaning pipeline to be executed again whenever updated data becomes available. This separation also simplifies debugging and future maintenance of the project.

The final stage of the ETL pipeline is the loading process, where the cleaned datasets are imported into a SQLite database. SQLite was selected because it is lightweight, serverless, and integrates seamlessly with Python, making it highly suitable for analytical projects. Each dataset is stored as an independent relational table, enabling efficient querying and reducing the need to repeatedly load large CSV files into memory. The use of a structured database also improves scalability and allows different analytical notebooks to access the same cleaned data without additional preprocessing. The database design follows a modular approach in which every table represents a specific business entity such as fund information, NAV history, investor transactions, portfolio holdings, or benchmark performance. The AMFI Code serves as the primary

identifier across multiple datasets, allowing relationships to be established between different tables during analysis. This relational structure makes it possible to perform complex joins, compare fund performance against benchmark indices, analyse investor behaviour, and evaluate portfolio composition efficiently.

Overall, the ETL pipeline and database architecture provide a strong technical foundation for the project. By automating data ingestion, ensuring high-quality preprocessing, and storing cleaned information within a relational database, the system creates a reliable environment for exploratory data analysis, financial performance evaluation, advanced analytics, and Power BI dashboard development.

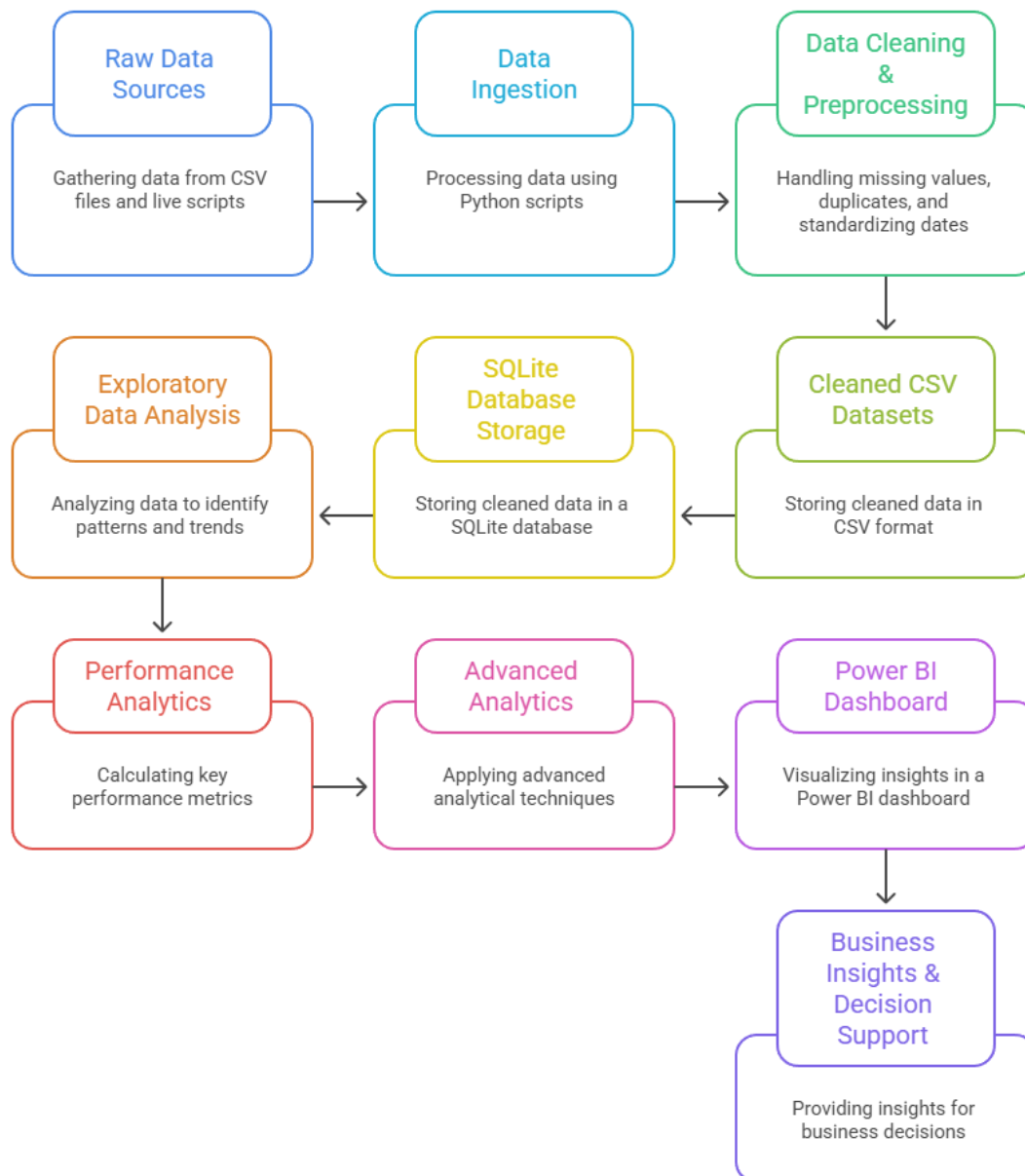


Fig: Workflow of the ETL Pipeline

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the mutual fund industry, identify investment trends, and discover meaningful relationships within the datasets before performing financial analysis. The cleaned datasets were analysed using Python libraries such as Pandas and Matplotlib to generate visualisations that summarize fund performance, investor behaviour, and industry growth. The insights obtained during this stage helped validate the quality of the data and guided the implementation of advanced financial metrics.

The analysis of **Assets Under Management (AUM)** reveals that a small number of fund houses dominate the Indian mutual fund industry, indicating a highly concentrated market where investors prefer established Asset Management Companies. This concentration reflects greater investor confidence in large AMCs and their ability to deliver consistent long-term performance.

Similarly, the **monthly SIP inflow trend** demonstrates a steady increase in systematic investments over time, highlighting the growing popularity of disciplined investing among retail investors. Rising SIP contributions indicate increasing financial awareness and a shift toward long-term wealth creation through mutual funds.

The **category-wise inflow heatmap** illustrates variations in investor preferences across different mutual fund categories. Equity-oriented funds consistently attract higher investments compared to debt-oriented categories, reflecting a stronger inclination toward long-term capital appreciation despite market volatility.

Correlation analysis of daily NAV returns shows that many schemes exhibit similar movement patterns because they are influenced by common market factors. However, the varying levels of correlation also suggest opportunities for diversification by combining funds with lower return correlations.

The distribution of expense ratios indicates that most schemes maintain competitive management costs, while only a limited number of funds charge relatively higher expenses. Since expense ratios directly influence long-term investment returns, this analysis provides useful information for investors when comparing similar schemes.

Overall, the Exploratory Data Analysis phase provides valuable insights into market structure, investor participation, fund characteristics, and portfolio behaviour. These findings establish the foundation for the performance evaluation and advanced analytical models presented in the following sections.

Performance Analytics and Advanced Analytics

The primary objective of this phase was to evaluate the performance of mutual fund schemes using industry-standard financial metrics while analysing the associated investment risks. Unlike traditional return analysis, which focuses only on historical growth, this project incorporates both return and risk-based indicators to provide a comprehensive assessment of fund performance. These metrics help investors compare schemes not only on profitability but also on consistency, volatility, and downside risk. The performance evaluation began with the calculation of **Compound Annual Growth Rate (CAGR)**, which measures the annualised return generated by a fund over a specific investment period. CAGR provides a standardised measure of growth and enables fair comparison among schemes with different investment horizons. To evaluate risk-adjusted performance, the **Sharpe Ratio** was computed for all mutual fund schemes by comparing excess returns against the overall volatility of returns. Funds with higher Sharpe Ratios demonstrate better compensation for the level of risk undertaken. Similarly, the **Sortino Ratio** was calculated to focus specifically on downside volatility, providing a more realistic assessment of investment risk by considering only negative return fluctuations.

The project further evaluates market sensitivity through **Alpha** and **Beta**. Alpha measures the excess return generated by a mutual fund relative to its benchmark after adjusting for market risk, while Beta indicates the sensitivity of fund returns to overall market movements. These metrics allow investors to identify funds capable of outperforming benchmark indices and understand how strongly individual schemes respond to market fluctuations. Additional indicators such as **Tracking Error** and **Maximum Drawdown** were also calculated to assess benchmark consistency and the maximum historical decline experienced by each scheme during adverse market conditions. To extend the analysis beyond conventional performance metrics, several advanced analytical techniques were implemented. **Historical Value at Risk (VaR)** and **Conditional Value at Risk (CVaR)** were calculated using the daily return distribution to estimate potential portfolio losses under adverse market conditions. While VaR represents the maximum expected loss at a 95% confidence level, CVaR measures the average loss beyond the VaR threshold, providing a more comprehensive assessment of downside risk.

The project also includes a **Rolling 90-Day Sharpe Ratio** to analyse how the risk-adjusted performance of selected mutual funds changes over time. Unlike a single Sharpe Ratio value, the rolling analysis captures periods of improving or deteriorating performance, allowing investors to understand the consistency of returns across different market environments. In addition, an **Investor Cohort Analysis** was performed by grouping investors according to their first investment year to identify long-term investment patterns and average contribution levels. A **SIP Continuity**

Analysis was conducted to evaluate the consistency of systematic investments, where investors exhibiting gaps greater than thirty-five days between SIP transactions were identified as potentially at-risk investors.

Finally, portfolio diversification was analysed using the **Herfindahl–Hirschman Index (HHI)**, which measures portfolio concentration based on sector allocation weights. Lower HHI values indicate better diversification, whereas higher values suggest greater concentration risk. A rule-based **Mutual Fund Recommendation System** was also developed to recommend suitable investment schemes according to an investor's risk appetite by combining risk category information with calculated Sharpe Ratios. Collectively, these analytical techniques provide a robust framework for evaluating mutual fund performance from both return and risk perspectives while supporting informed investment decision-making.

Power BI Dashboard

To transform analytical findings into an interactive and user-friendly format, a comprehensive dashboard was developed using Microsoft Power BI. The dashboard was designed to provide stakeholders with a clear overview of mutual fund performance, investor behaviour, and industry trends through dynamic visualisations and interactive filters. Rather than presenting static reports, the dashboard enables users to explore data from multiple perspectives using slicers, KPI cards, and comparative charts. The dashboard is organised into four dedicated pages, each focusing on a specific aspect of the analysis. The Industry Overview page presents high-level insights into the mutual fund sector, including total Assets Under Management (AUM), monthly SIP inflows, total folio count, and the number of active mutual fund schemes. These KPIs provide a quick summary of the industry's growth and current market position. The Fund Performance page focuses on evaluating individual mutual fund schemes using financial performance metrics such as Sharpe Ratio, Fund Score, Alpha, Beta, and other risk-adjusted indicators. Interactive charts allow users to compare funds based on category, benchmark, and risk profile, helping identify schemes that consistently deliver strong performance while maintaining acceptable levels of risk. The Investor Analytics page examines investor behaviour through demographic analysis, investment patterns, and geographical distribution. Information such as age groups, gender, city tiers, and investment preferences helps understand how different investor segments participate in the mutual fund ecosystem. This analysis provides valuable insights into customer behaviour and highlights emerging investment trends. The SIP and Market Trends page visualises the relationship between systematic investments and market performance. Trend charts display SIP inflows over time, benchmark index movement, category-wise investments, and other market indicators that assist in understanding long-term investment behaviour and overall industry growth. To improve usability, the dashboard incorporates interactive slicers that allow users to filter data by mutual fund category, fund house, benchmark index, and risk category. This functionality enables decision-makers to customise the analysis according to their requirements without modifying the underlying data model. The combination of KPI cards, scorecards, bar charts, line charts, heatmaps, and trend visualisations provides an intuitive interface that supports efficient data exploration and informed investment decision-making.

Overall, the Power BI dashboard serves as the final presentation layer of the project by converting complex financial analyses into meaningful and easily interpretable business insights. It demonstrates how business intelligence tools can effectively communicate analytical findings and support strategic decision-making in the financial services domain.

Mutual Fund Industry Dashboard

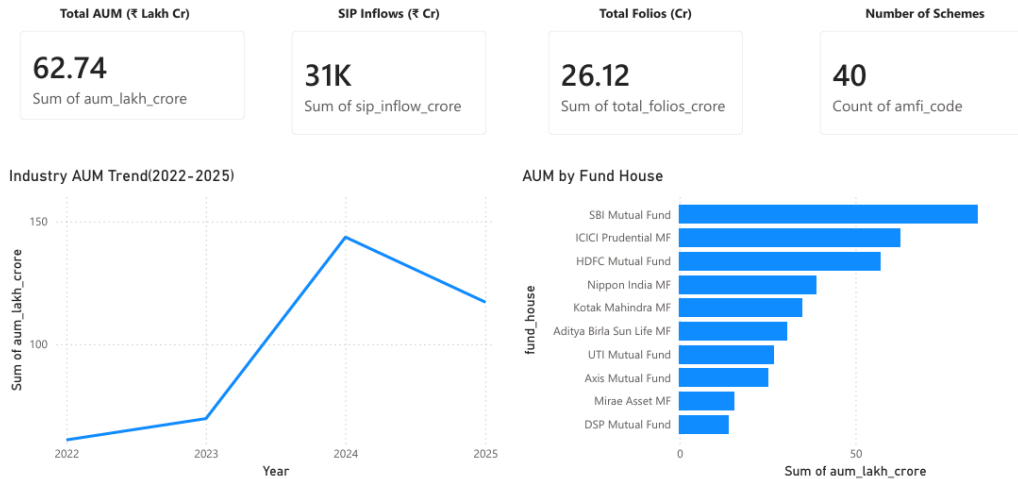


fig:Mutual Fund Industry Overview

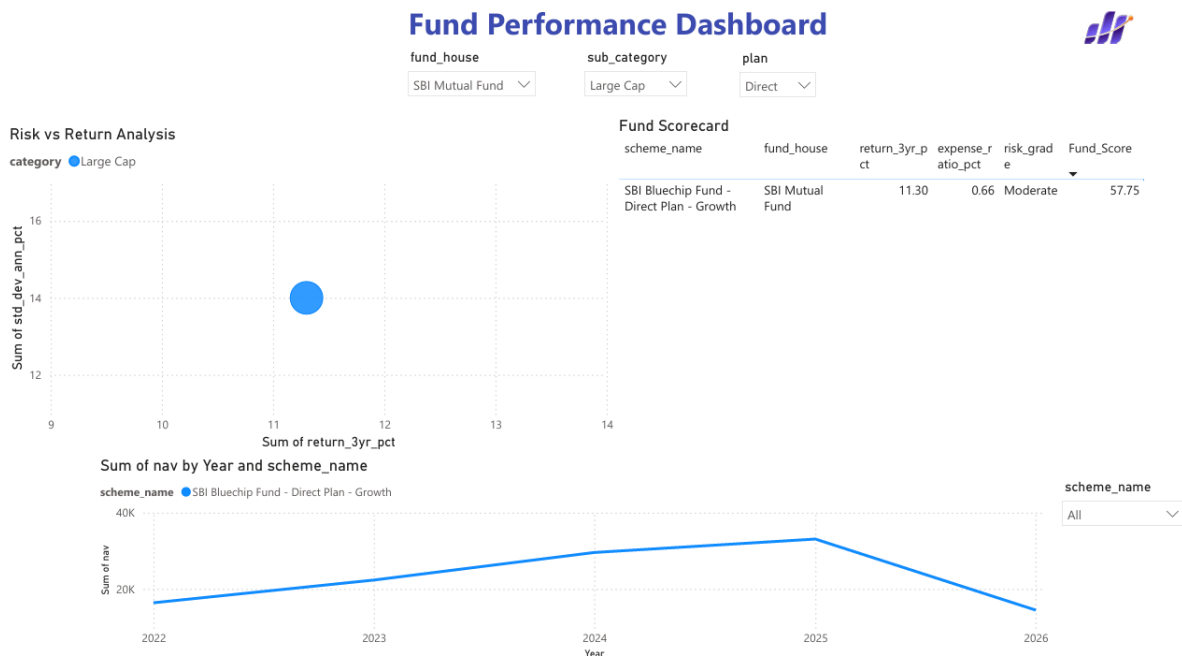


fig:Fund Performance Analysis

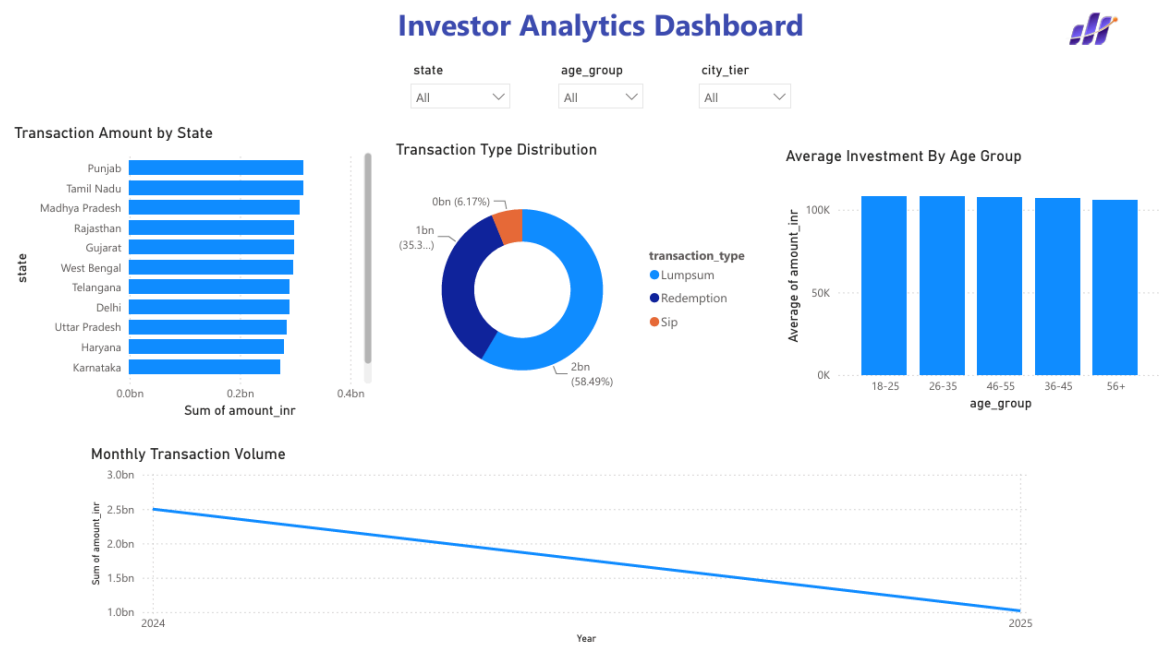


Fig: Investor Analytics

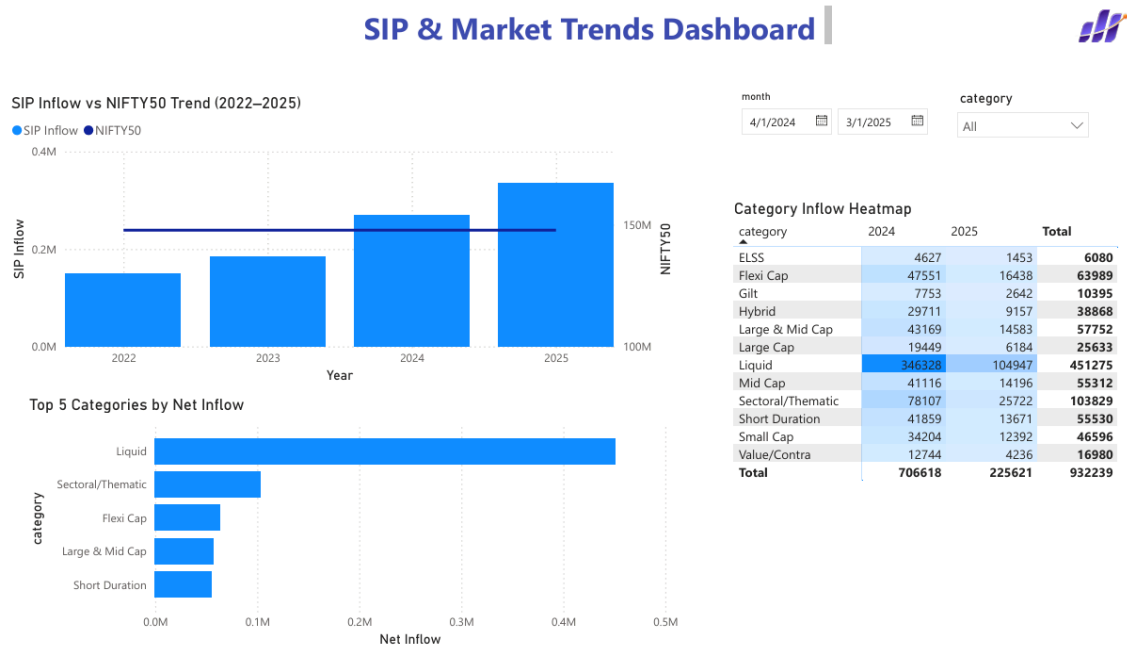


fig:SIP&Market trends

Key Findings

The project successfully transformed raw mutual fund datasets into a structured analytical platform capable of evaluating fund performance, investor behaviour, and market trends. Analysis of industry data revealed that a small number of Asset Management Companies dominate the market in terms of Assets Under Management, while Systematic Investment Plan (SIP) inflows have shown consistent growth, reflecting increasing investor participation and financial awareness. Risk-adjusted performance metrics such as Sharpe Ratio, Sortino Ratio, Alpha, and Beta enabled effective comparison between mutual fund schemes beyond traditional return analysis. Advanced analytics including Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, and portfolio concentration using the Herfindahl–Hirschman Index provided deeper insights into investment risk, portfolio diversification, and investor behaviour. The interactive Power BI dashboard successfully consolidated these insights into an intuitive decision-support system for analysing industry performance and mutual fund trends.

Challenges Faced

During the implementation of this project, several challenges were encountered. The datasets contained inconsistencies such as missing values, duplicate records, varying date formats, and inconsistent column names, which required extensive preprocessing before analysis. Integrating multiple datasets while maintaining relational consistency across tables was another significant challenge. Designing meaningful financial metrics and ensuring accurate implementation of statistical formulas required careful validation. Developing an interactive dashboard that effectively balanced analytical depth with visual simplicity also required multiple iterations to achieve a user-friendly interface.

Future Scope

The current system can be extended by integrating live NAV data through financial APIs to provide real-time fund performance analysis. Machine learning models may be incorporated to forecast NAV movements and classify mutual fund risk levels. Portfolio optimisation techniques such as the Markowitz Efficient Frontier can assist investors in constructing optimal investment portfolios. The solution can also be deployed as a web application using Streamlit or integrated with cloud-based databases to improve scalability, accessibility, and real-time reporting capabilities.

Conclusion

The **Mutual Fund Analytics & Performance Evaluation System** demonstrates how data engineering, financial analytics, and business intelligence can be integrated into a single analytical platform. Beginning with data ingestion and preprocessing, the project successfully implemented an automated ETL pipeline, structured SQLite database, comprehensive financial performance analysis, advanced analytical models, and an interactive Power BI dashboard. The resulting system provides meaningful insights into mutual fund performance, investor behaviour, and market trends while supporting informed investment decisions. Overall, the project highlights the practical application of Python, SQL, Power BI, and data analytics techniques in solving real-world financial problems and serves as a scalable foundation for future enhancements in investment analytics.